Task 2 — Datasheet Literacy

Selected Component

- **Component Name:** Motor Driver TB6612FNG Module (Dual DC Motor Driver)
- **Supplier Link**: Robu.in Product Page
- **Key Specifications:** Dual DC motor control, PWM speed regulation, 1.2 A average continuous current per channel (3.2 A peak), efficient MOSFET-based H-bridge design.

1. Datasheet Parameters Constraining Circuit Design

The datasheet specifications of the TB6612FNG directly dictated several key hardware and wiring decisions in our robot design:

- Motor Supply Voltage Range (V_M: 2.5V to 13.5V operating, 15V max)
    - *Constraint & Influence*: The V_M pin powers the motor H-bridge. To ensure our N20 gear motors receive sufficient torque without exceeding voltage limits, we selected a Bonka 7.4V 2200mAh 2S LiPo battery pack in our BOM. Its nominal 7.4V (and ~8.4V fully charged) output sits comfortably within the 2.5V–13.5V operating window, avoiding both under-voltage sluggishness and the 15V absolute maximum voltage breakdown limit.

- Logic Supply Voltage Range (V_{CC}: 2.7V to 5.5V)
    - *Constraint & Influence*: The V_{CC} pin powers the internal logic gates and level shifters. Because our main controller is an ESP-WROOM-32 operating on 3.3V logic, we wired the driver's V_{CC} pin to the 3.3V regulated rail.

      Furthermore, the control inputs (AIN1, AIN2, BIN1, BIN2, PWMA, PWMB, STBY) accept logic high levels up to 5.5V, ensuring direct, seamless compatibility with the ESP32 GPIO pins without requiring external logic level converters.

- Continuous Output Current (I_{out}: 1.2 A average per channel)
    - *Constraint & Influence*: The GA12-N20 6V micro DC motors draw low operating currents (typically well under 1.0 A even under load). Knowing the continuous current limit is 1.2 A per channel gave us confidence that the TB6612FNG can drive both motors during differential steering and line tracking without entering thermal overload or tripping the built-in thermal shutdown protection.

2. Component Damage Condition & Avoidance Strategy

- Critical Hazard: Flyback Voltage Spike (Back-EMF) and exceeding the Output Absolute Maximum Voltage rating ($V_{out} \leq 15V$).
    1. *Mechanism of Damage*: When DC motors rapidly switch direction, brake (short brake mode), or decelerate via PWM, the collapsing magnetic fields in the motor windings generate sharp inductive kickback voltage spikes (back-EMF). If unmitigated, these transient spikes can exceed the 15V absolute maximum rating and puncture the internal LD-MOS H-bridge output transistors.

- Design & Wiring Avoidance Measures:
    1. Integrated H-Bridge Clamping Diodes: The TB6612FNG incorporates internal flyback/freewheeling diodes connected across its output stages to safely clamp inductive kickback currents back to the V_M power rail and ground.
    2. Bulk Decoupling Capacitance: We included power decoupling capacitors across the V_M and ground input rails close to the driver module to absorb high-frequency transient voltage ripples and stabilize the supply during rapid PWM transitions.
    3. Software Control & Dead-Time Management: The firmware avoids abrupt, aggressive hard-reversals of the motors when possible. Additionally, the driver IC features built-in internal dead-time control (t_2, t_4) during switching states to prevent shoot-through currents.